

Cardiff School of Technologies

Assessment Brief

Module Code

DAT5004

Module Title

Big Data Management Systems

Academic Year

2024-25

Semester

II

Module Leader email

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Assessment Details

Assessment title	Abr.	Weighting
Web scraping and database implementation	PRAC1	50%
Pass marks are 40% for undergraduate work and 50% for postgraduate work unless stated otherwise.		

Task/assessment brief:

Task 1 (Web scraping)

Web Scraping Task for Extracting Movie Information from FilmCrave

FilmCrave (https://www.filmcrave.com/list_top_movie.php) is a website where people share their favourite movies, reviews, and ratings. It has a huge collection of films, complete with rankings, genres, and other details. The goal of this project is to gather information on over 1,000 movies from this site. The data we want includes the Year of Release, Overall Rating, Language, Genre, MPAA Rating (like PG or R), Director, Actors, and a short Plot summary.

To do this, you'll use Python. A tool called BeautifulSoup will help you read and understand the webpage's layout to pull out the data we need. Another tool, Requests, will help you connect to the website and fetch its content. Once you have the data, you'll use Pandas to organise it neatly into a table format, and then save it as a CSV file.

After gathering the data, you'll clean it up, fix any missing or incorrect values, and make sure everything is consistent. Then, you'll analyse it. you'll create visual charts, like bar graphs showing popular genres or line graphs tracking how movie ratings have changed over time, using tools like Matplotlib and Seaborn. This project will teach valuable skills in web scraping, data organisation, and analysis.

You need to provide the solutions to Task 1 using a single .ipynb/.py file.

Task 2 (SQLite application)

In this task, you are required to develop a Python application to be used for storing the data under a relational database architecture. We are interested about the 'International Cricket Database Management System', where we can store the details of cricketers, find the particular cricketer with different attributes, view all records, etc. The recommended Python library for the SQL part of this application is sqlite3. Upon execution, the application should allow the user to choose one of the following seven options: (Please note that the user interface might appear differently in your case, depending upon the tool and platform chosen by you for implementation. All outputs should be displayed in the form of table)

Menu after Each Operation:

After every operation, such as adding, updating, or viewing cricketers, the following menu should appear for the user to make another selection:

International Cricket Database

1. Insert Cricketer
2. View All Cricketers
3. Find Cricketer by Name
4. Find Best Score
5. Update Cricketer
6. Delete Cricketer
7. Exit

Enter your choice:

Sample Example:

1. Insert a New Cricketer

○ Input:

1

Enter cricketers ID: 6
Enter cricketer's name: MS Dhoni
Enter country: India
Enter matches played: 350
Enter runs scored: 10773
Enter highest score: 183
Enter batting average: 51.34

○ Output:

Cricketer added successfully.

Likewise, insert records for other cricketers.

2. View All Cricketers

○ Input:

2

○ Output (that output may be in different format in your case):

Stored Cricketers:
ID: 1, Name: Sachin Tendulkar, Country: India, Matches: 463, Runs: 18426, Highest Score: 200, Average: 44.83
ID: 2, Name: Virat Kohli, Country: India, Matches: 254, Runs: 12169, Highest Score: 183, Average: 59.07
ID: 3, Name: Ricky Ponting, Country: Australia, Matches: 375, Runs: 13704, Highest Score: 164, Average: 51.85
ID: 4, Name: Brian Lara, Country: West Indies, Matches: 131, Runs: 11953, Highest Score: 400, Average: 52.88
ID: 5, Name: Jacques Kallis, Country: South Africa, Matches: 166, Runs: 10294, Highest Score: 189, Average: 45.97
ID: 6, Name: MS Dhoni, Country: India, Matches: 350, Runs: 10773, Highest Score: 183, Average: 51.34

3. Find Cricketer by Name

○ Input:

Enter the cricketer's name to search: Virat

○ Output:

ID: 2, Name: Virat Kohli, Country: India, Matches: 254, Runs: 12169, Highest Score: 183, Average: 59.07

4. Find Best Score

○ Input:

4

- **Output (that output may be in different format in your case):**

Cricketer with the highest score:

Name: Brian Lara, Country: West Indies, Highest Score: 400, Matches: 131, Runs: 11953, Average: 52.88

5. Update a Cricketer's Record

- **Input:**

5

Enter the ID of the cricketer to update: 6

Which field would you like to update? (name/country/matches/runs/highest_score/average): runs

Enter the new value for runs: 11000

- **Output:**

Cricketer updated successfully.

6. Delete a Cricketer

- **Input:**

6

Enter the ID of the cricketer to delete: 6

- **Output:**

Cricketer deleted successfully.

7. Exit

- **Input:**

7

- **Output:**

Exiting the application.

NOTE: We will go through a few related examples of Python-based applications relevant to relational database during the workshop which will be helpful in your planning and execution for this task.

You need to provide the solutions to Task 2 using a Python project (collection of .py/.ipynb files).

Task 3 (MongoDB):

In this task, you need to work with MongoDB Atlas, a cloud version of MongoDB to perform the CRUD operations. The Python-based driver PyMongo will be needed to interact with the cluster created by you on MongoDB Atlas. As a starting point, you will be required to use a Titanic dataset (Titanic_data.csv), available on Moodle. This task can be completed by the following subtasks:

Task 3.1 (MongoDB setup):

Start by creating a free cluster on MongoDB Atlas, such as *Cluster 0*, or use the one created during the workshop sessions, as MongoDB typically allows only one free cluster per user. Use a Jupyter notebook (.ipynb file) to write and present your Python code. Connect to the cluster using the PyMongo library, which acts as the driver for interacting with MongoDB. Load the Titanic dataset (Titanic_data.csv) into a Pandas DataFrame for easy handling and analysis. Then, use PyMongo to transfer the data from the DataFrame into a database and collection in your MongoDB cluster.

You need to provide the solution to Task 3.1 using a screenshot of your working cluster on MongoDB Atlas.

Task 3.2 (Loading Data):

Create a database and a collection within the current cluster of your MongoDB Atlas. You need to store the Titanic data in the collection. Using suitable Python codes, copy the data from Titanic DataFrame into the newly created collection within your database.

Assume that the original creator of the Titanic data added five extra rows with incomplete information to the Titanic_data.csv file. For example, the new rows are missing details such as 'Name,' 'PassengerId,' and 'Embarked.' These inconsistencies could complicate data storage in a Relational Database, which requires strict adherence to schemas. However, MongoDB's flexible schema allows you to handle this seamlessly. Insert five additional documents into your existing collection, make sure that only the available fields are populated. Feel free to use dummy data to supplement missing information and make certain your collection remains functional for analysis.

Task 3.3 (MongoDB query):

Task 3.3.1

Write code to count the number of passengers who survived or did not survive, grouped by passenger class (Pclass).

Task 3.3.2

Write code to find the average age of survivors and non-survivors, grouped by passenger class (Pclass).

Task 3.3.3

Write a code to find the total fare paid by survivors and non-survivors, grouped by gender (Sex).

Task 3.3.4

Write a code to find the percentage of survivors in each class (Pclass).

Task 3.3.5

Write code to find the passengers who paid the highest fare, along with their details.

Task 3.3.6

Write code to find the number of missing values for each field (e.g., Age, Embarked, etc.).

Task 3.3.7

Write a code to find passengers who survived and have missing Age information.

Task 3.3.8

Write a code to find the number of passengers who survived, grouped by age group (e.g., 0-10, 11-20, etc.).

Task 3.3.9

Find the number of passengers who survived, grouped by the first letter of their name.

Task 3.3.10

Write a code to find the median fare paid by passengers in each class (Pclass).

Task 3.3.11

Write a code to find the passengers who were in the same family (same surname).

Task 3.3.12

Write a code to find the survival rate for male and female passengers.

You need to provide the solutions to Tasks 3.2 and 3.3 using a single .ipynb/.py file.

CODE PROFESSIONALISM:

You are required to present all the codes in a professional manner. This broadly implies that all the functions and major steps within the Python script should be explained satisfactorily. The code must be cross-compatible and not specific to a specific operating system.

Word count (or equivalent):

2000

This is a reflection of the effort required for the assessment. Word counts will normally include any text, tables, calculations, figures, subtitles and citations. Reference lists and contents of appendices are excluded from the word count. Contents of appendices are not usually considered when determining your final assessment grade.

Academic or technical terms explained:

Web scraping – Web scraping is the process of automatically extracting data from websites. It involves writing code or using specialised software tools to scrape data from a website's HTML code or its Application Programming Interface (API).

SQLite – SQLite is a relational database management system that is designed to be embedded into applications. It is a software library that provides a lightweight disk-based database that does not require a separate server process and allows for simple and efficient data storage and retrieval.

MongoDB – MongoDB is a document-oriented, NoSQL database that is designed to store and manage unstructured data. Unlike traditional relational databases, which organize data into tables with fixed schemas, MongoDB stores data in flexible, JSON-like documents with dynamic schemas.

CRUD – Create, Read, Update, Delete

NoSQL – NoSQL stands for "not only SQL" and is a broad category of database management systems that are designed to handle unstructured, semi-structured, or highly variable data. Unlike traditional relational databases, which store data in fixed tables with predefined schemas, NoSQL databases use a variety of data models and structures, such as key-value stores, document-oriented databases, graph databases, and column-family stores.

Key Bloom elements:

Comprehension - Understand key concepts: web scraping, relational databases, and MongoDB operations for data extraction, storage, and manipulation.

Analysis - Decompose tasks into components: data fetching, cleaning, storing, and querying, aligning with the tools used (SQLite, MongoDB, etc.).

Evaluation - Compare relational (SQLite) and NoSQL (MongoDB) database approaches for handling structured and unstructured data.

Synthesis - Aggregating the intermediate results of web scraping into a sharable data format like Pandas DataFrame; Similarly synthesising the query results of MongoDB to provide an overview of a typical document-based NoSQL database system.

Application - Python-based database management tools such as, sqlite3 and PyMongo as well as application of Python in web scraping using Beautiful Library soup.

Artificial Intelligence Models – Guidance for this assessment:

Artificial Intelligence (AI) models can be a powerful tool to support your learning. The University has provided some resources to support you in its appropriate usage:

- [Library Services AI Hub](#)
- [Student Guide to AI and Assessment](#)
- [Code of Conduct for Students on the use of AI](#)
- [Cite Them Right resource on citing materials relating to AI \(if permitted\)](#)

As per the academic regulations ([Academic Handbook Ah1_08](#)), in all cases you must submit work that is your own, acknowledging any part of it that has been informed by another source – including that which is AI generated. Upon submission of work, you will be asked to confirm the following statement:

I confirm that this assignment is my own work, except where I have acknowledged the use of works from other sources, including the use of any artificial intelligence (AI) tools, in accordance with what is allowable as described in the assessment brief.

Please note the following:

- AI should not be used as a substitute for your own knowledge, and you should never include any material that you do not understand and could not explain if asked.
- Not being able to explain your work when asked is likely to be a key factor when considering cases of academic misconduct related to AI.

The following information provides specific guidance for this assessment about what level of AI use is appropriate for this assessment. Remember that in all cases you must submit work that is your own, acknowledging any part of it that has been provided by another source.

NO USE OF GENERATIVE AI EXPECTED • Your assignment should be produced using information sourced by you from your learning materials and academic sources and cited appropriately. • AI tools for checking spelling, grammar and referencing may be used.	
AI ACKNOWLEDGED • You can use AI tools to learn about your topic, as part of your study, or in preparing initial guidance on assignments (e.g. headline structure, suggestions for inclusion of topics). • Any materials that you have sourced from AI should be rewritten or reconfigured and integrated into your own work and referenced appropriately. It is recommended that this is confirmed by a relevant academic source. • Any support gained from AI should be acknowledged in a statement at the end of the assignment, making clear what the support was, and how you used it and developed it for your own work. Example statements are available in the Student Code of Conduct [link].	
AI EMBEDDED • Use of AI is an integral and expected part of the assessment. • The explicit inclusion of AI within the assessment means that instructions on the expected use will be part of the assessment brief. • Your assessment brief will describe how you should acknowledge the way in which you used AI tools.	

Submission Details

Submission Deadline:	13 June 2025	Estimated Feedback Return Date	This will normally be 20 working days after initial submission.
Submission Time:	By 4.00pm on the deadline day.		
Moodle/Turnitin:	Any assessments submitted after the deadline will not be marked and will be recorded as a non-attempt unless you have had an extension request agreed or have approved mitigating circumstances. See the School Moodle pages for more information on extensions and mitigating circumstances.		

File Format:

The assessment **MUST** be submitted through the Turnitin submission point in Moodle. All the Python files such as, .py/.ipynb, any miscellaneous files required for running all the tasks, and a READ.txt (brief manual for running your programmes) must be submitted individually.

Your assessment should be titled with your Student ID Number, module code and assessment id, e.g., st12345678 DAT5004 PRAC1

Feedback

Feedback for the assessment will be provided electronically via Moodle. Feedback will be provided with comments on your strengths and the areas which you can improve. View the [guidance](#) on how to access your feedback.

All marks are provisional and are subject to [quality assurance processes](#) and confirmation at the programme Examination Board.

Assessment Criteria

Learning outcomes assessed

Learning Outcomes	Assessment Criteria
[LO1] Discuss, analyse and apply techniques for information, data and knowledge management.	[AC1] Task 1 (LO1) Web scraping
[LO2] Analyse and explain the implications of technological innovation for data usage, data security and data management.	[AC2] Task 2 (LO3) SQLite-based application
[LO3] Reflect on the limitations of knowledge management systems and technologies in a socio-technical and organisational context.	[AC3] Task 3.1 (LO2) MongoDB set-up
	[AC4] Task 3.2 (LO2) Loading data in MongoDB
	[AC5] Task 3.3 (LO2) Queries in MongoDB
	[AC6] Code professionalism (Applicable to all the tasks; Task 1, Task 2, and Task 3)

Other skills/attributes developed

This includes elements of the Cardiff Met EDGE (Ethical, Digital, Global and Entrepreneurial skills) and other attributes developed in students through the completion of the module and assessment. These will also be highlighted in the module guidance, which should be read by all students completing the module. Assessments are not just a way of auditing student knowledge. They are a process which provides additional learning and development through the preparation for and completion of the assessment.

ETHICAL	Critical understanding of the importance of ethical, social, and legal agendas in Big Data Management Systems.
DIGITAL	Advanced digital skills for the fourth industrial revolution: web scraping, data visualisation and non-relational database management system.
GLOBAL	International horizon and benchmarking for Big Data Management System, corporate specialist software (MongoDB) and Python programming knowledge, and other third-party tool.
ENTREPRENEURIAL	Use students' creativity to solve database management system related problem with the help of Entrepreneurship related software MangoDB and Programming language Python.

Marking/Assessment Criteria

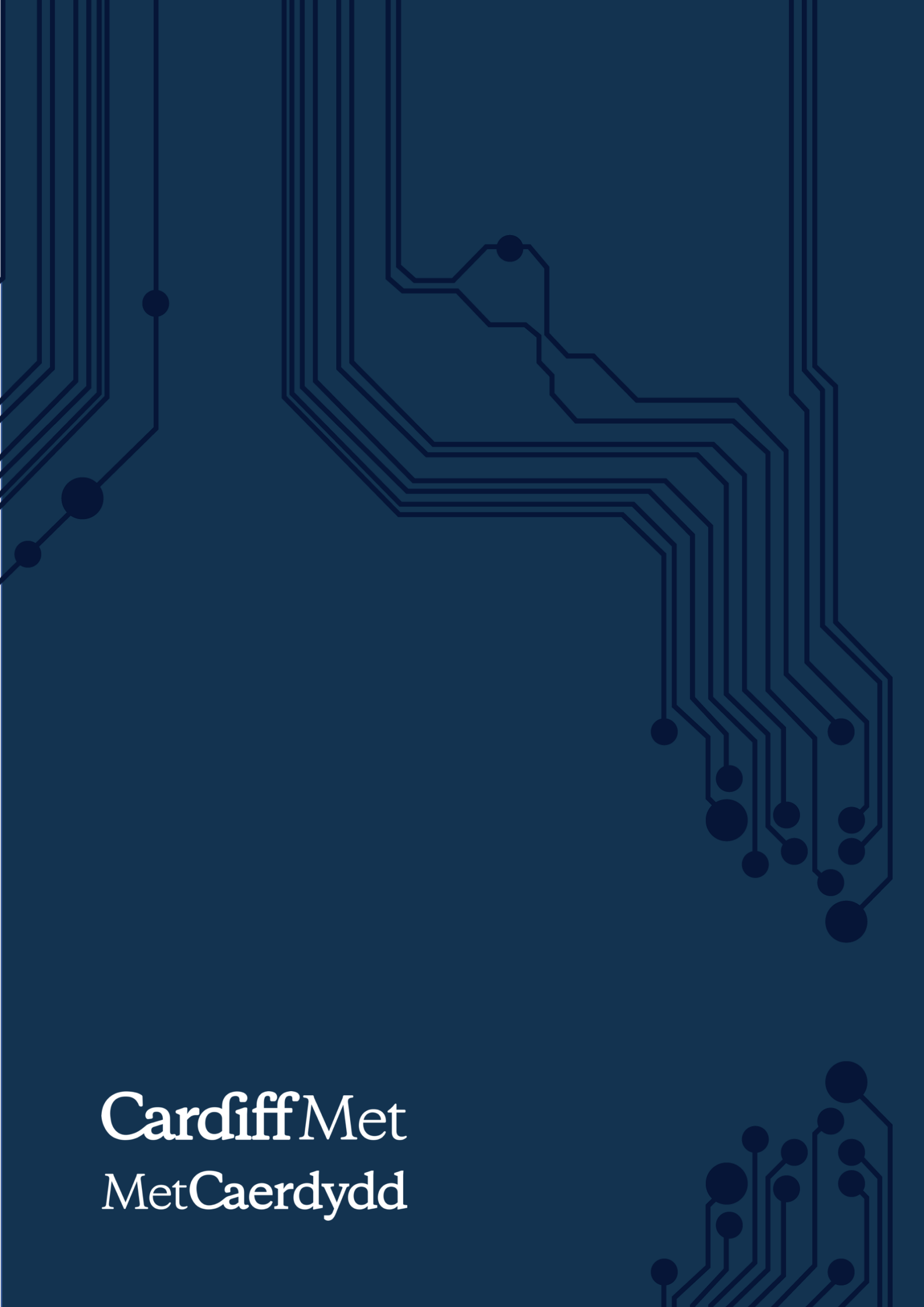
Assessment Criteria	100%
Task 1 (LO1) Web scraping	25%
Task 2 (LO3) SQLite-based application	35%
Task 3.1 (LO2) MongoDB set-up	6%
Task 3.2 (LO2) Loading data in MongoDB	5%
Task 3.3 (LO2) Queries in MongoDB	24%
Code professionalism (Applicable to all the tasks; Task 1, Task 2, and Task 3)	5%

Further Information on assessment, referencing and grading can be found in the Module Handbook (on Moodle)

Marking/Assessment Criteria

Tasks	1-29% (Clear Fail)	30-39% (Narrow Fail)	40-49% (3rd)	50-59% (2:2)	60-69% (2:1)	70-79% (1st)	80 – 100% (1 st class+)
Task 1	Very poor, incomplete and/or irrelevant. Demonstrates a serious lack of comprehension and/or engagement with the set task. May have misunderstood the set task. No learning outcomes are met in full, although there may be minimal attainment in one or two areas.	Limited/no progress towards a solution to the specified web scraping problem. Unprofessional solution. Does not demonstrate understanding of the problem and/or solution.	An unfocused and unspecific solution to web scraping. Contain some error but there is evidence to show that some effort has been spent in the web scraping	Substantive, but a non-specific attempt at web scraping. Contain some errors, but there is evidence to show a somewhat satisfactory amount of effort has been spent in following the steps of web scraping	A substantive and focused attempt at web scraping. Clear degree of understanding of the given problem. Contains a few errors, appropriate for the amount of time and effort spent in the web scraping.	A substantive and specific solution to web scraping. Demonstrates a high degree of understanding of the steps of web scraping. Contains very few minor errors.	A specific and substantive solution to web scraping. The approach is rigorous, interesting and has significant practical importance. Demonstrate an excellent understanding of the given problem. Error-free content.
Task 2	very poor communication of information and ideas. Very poor presentation and inappropriate format. Contain serious/numerous errors and misunderstandings. No learning outcomes are met in full. Demonstrates a serious lack of comprehension and/or engagement	Limited/no progress towards a solution to the specified SQLite-based application development. Unprofessional solution. Does not demonstrate understanding of the problem and/or solution.	An unfocused and unspecific solution to the specified SQLite-based application development. Contain some error but there is evidence to show that some effort has been spent in SQLite-based application development.	Substantive, but a non-specific attempt at SQLite-based application development. Contain some errors, but there is evidence to show a somewhat satisfactory amount of effort has been spent in following the steps of application development.	A substantive and focused attempt at SQLite-based application development. Clear degree of understanding of the given problem. Contains a few errors, appropriate for the amount of time and effort spent in the application development.	A substantive and specific solution to SQLite-based application development. Demonstrates a high degree of understanding of the steps of application development. Contains very few minor errors.	A specific and substantive solution to SQLite-based application development. The approach is rigorous, interesting and has significant practical importance. Demonstrate an excellent understanding of the given problem. Error-free content.

	with the set task. May have misunderstood the set task.						
Task 3	Very poor, incomplete and/or irrelevant. Demonstrates a serious lack of comprehension and/or engagement with the set task. May have misunderstood the set task. No learning outcomes are met in full, although there may be minimal attainment in one or two areas.	Limited/no progress towards a solution to the specified MongoDB set-up and associated queries. Unprofessional solution. Does not demonstrate understanding of the problem and/or solution.	An unfocused and unspecific solution to the specified MongoDB set-up and associated queries. Contain some error but there is evidence to show that some effort has been spent in MongoDB set-up and associated queries.	Substantive, but a non-specific attempt at MongoDB set-up and associated queries. Contain some errors, but there is evidence to show a somewhat satisfactory amount of effort has been spent in following the steps of MongoDB set-up and associated queries.	A substantive and focused attempt at MongoDB set-up and associated queries. Clear degree of understanding of the given problem. Contains a few errors, appropriate for the amount of time and effort spent in MongoDB set-up and associated queries	A substantive and specific solution to MongoDB set-up and associated queries. Demonstrates a high degree of understanding of the steps of MongoDB set-up and associated queries. Contains very few minor errors.	A specific and substantive solution to MongoDB set-up and associated queries. The approach is rigorous, interesting and has significant practical importance. Demonstrate an excellent understanding of the given problem. Error-free content.

An abstract graphic design featuring a dark blue background with a complex, light blue circuit-like pattern. The pattern consists of numerous thin, parallel lines that branch out and connect to various circular nodes of different sizes. The lines and nodes are arranged in a way that suggests a network or a digital circuit, with some lines running vertically and others branching out horizontally or diagonally. The overall effect is a sense of connectivity and technology.

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